

printenv | grep HOME

printenv | grep USER

```
saira@ubuntu:~$ printenv | grep SHELL
SHELL=/bin/bash
saira@ubuntu:~$ printenv | grep HOME
HOME=/home/saira
saira@ubuntu:~$ printenv | grep USER
USER=saira
saira@ubuntu:~$ _
```

Task 2 — Export DB_* variables temporarily and observe scope

Goal: Create env variables with export in the current shell, verify them, then close shell and show variables are gone.

Steps:

1. Define all DB_* variables (run the three exports one after another). Capture them in one screenshot showing the three export commands and their execution:

export DB_URL="postgres://db.example.local:5432/mydb"

export DB_USER="labuser"

export DB_PASSWORD="labpass123"

```
saira@ubuntu:~$ export DB_URL="postgres://db.example.local:5432/mydb"
saira@ubuntu:~$ export DB_USER="labuser"
saira@ubuntu:~$ export DB_PASSWORD="labpass123"
saira@ubuntu:~$
```

2. Echo the three variables (run the three echo commands together) and capture one screenshot showing their outputs:

echo "\$DB_URL"

echo "\$DB_USER"

echo "\$DB_PASSWORD"

```
saira@ubuntu:~$ echo "$DB_URL"
postgres://db.example.local:5432/mydb
saira@ubuntu:~$ echo "$DB_USER"
labuser
saira@ubuntu:~$ echo "$DB_PASSWORD"
labpass123
saira@ubuntu:~$
```

3. Show all DB_ variables with a single grep command (capture that output):

printenv | grep '^DB_'

```
saira@ubuntu:~$ printenv | grep '^DB_'
DB_PASSWORD=labpass123
DB_USER=labuser
DB_URL=postgres://db.example.local:5432/mydb
saira@ubuntu:~$ _
```

4. Close the bash session (e.g., exit) and reopen a new terminal. Verify the variables are gone by running the echo(s) and the grep together; capture both checks in one screenshot:

echo "\$DB_URL"

printenv | grep '^DB_'

```
saira@ubuntu:~$ echo "$DB_URL"
postgres://db.example.local:5432/mydb
saira@ubuntu:~$ printenv | grep '^DB_'
DB_PASSWORD=labpass123
DB_USER=labuser
DB_URL=postgres://db.example.local:5432/mydb
saira@ubuntu:~$
```

Optional combined evidence (if you want to also save a combined screenshot showing steps 1–3 together for convenience):

```
saira@ubuntu:~$ export DB_URL="postgres://db.example.local:5432/mydb"
saira@ubuntu:~$ export DB_USER="labuser"
saira@ubuntu:~$ export DB_PASSWORD="labpass123"
saira@ubuntu:~$ echo "$DB_URL"
postgres://db.example.local:5432/mydb
saira@ubuntu:~$ echo "$DB_USER"
labuser
saira@ubuntu:~$ echo "$DB_PASSWORD"
labpass123
saira@ubuntu:~$ printenv | grep '^DB_'
DB_PASSWORD=labpass123
DB_USER=labuser
DB_URL=postgres://db.example.local:5432/mydb
saira@ubuntu:~$
```

Task 3 — Make DB_* variables persistent in ~/.bashrc

Goal: Add DB_* variables to ~/.bashrc, reload, and verify persistence. Grouped captures:

show the three export lines in ~/.bashrc together, and group the post-source checks into one screenshot.

Steps:

1. Open ~/.bashrc in an editor and append the three export lines. Capture the editor showing the three lines added (single screenshot):

vim ~/.bashrc

add at the end:

Lab 7 persistent DB variables

export DB_URL="postgres://db.example.local:5432/mydb"

export DB_USER="labuser"

export DB_PASSWORD="labpass123"

```

#alias dir='dir' --color=auto'
#alias vdir='vdir' --color=auto'

alias grep='grep --color=auto'
alias fgrep='fgrep --color=auto'
alias egrep='egrep --color=auto'
fi

# colored GCC warnings and errors
#export GCC_COLORS='error=01;31:warning=01;35:note=01;36:caret=01;32:locus=01:quote=01'

# some more ls aliases
alias ll='ls -alF'
alias la='ls -A'
alias l='ls -CF'

# Add an "alert" alias for long running commands. Use like so:
# sleep 10; alert
alias alert='notify-send --urgency=low -i "${?} = 0" && echo terminal || echo error)' "${history|tail -n1|sed -e 's/\s*[0-9]\+\s*//;s/[:&]\s*alert$/\s*'
)"

# Alias definitions.
# You may want to put all your additions into a separate file like
# ~/.bash_aliases, instead of adding them here directly.
# See /usr/share/doc/bash-doc/examples in the bash-doc package.

if [ -f ~/.bash_aliases ]; then
    . ~/.bash_aliases
fi

# enable programmable completion features (you don't need to enable
# this, if it's already enabled in /etc/bash.bashrc and /etc/profile
# sources /etc/bash.bashrc).
if ! shopt -o posix; then
    if [ -f /usr/share/bash-completion/bash_completion ]; then
        . /usr/share/bash-completion/bash_completion
    elif [ -f /etc/bash_completion ]; then
        . /etc/bash_completion
    fi
fi

# add at the end:
# Lab 7 persistent DB variables

export DB_URL="postgres://db.example.local:5432/mydb"
export DB_USER="labuser"
export DB_PASSWORD="labpass123"

-- INSERT --

```

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2. Source ~/.bashrc and capture the source command in one screenshot together with the next verification commands (grouped): run source ~/.bashrc and then immediately run the three echoes and a single grep, capturing all of these in one screenshot:

source ~/.bashrc

echo "\$DB_URL"

echo "\$DB_USER"

echo "\$DB_PASSWORD"

printenv | grep '^DB_'

```

saira@ubuntu:~$ source ~/.bashrc
saira@ubuntu:~$ echo "$DB_URL"
postgres://db.example.local:5432/mydb
saira@ubuntu:~$
saira@ubuntu:~$
saira@ubuntu:~$ source ~/.bashrc
saira@ubuntu:~$ echo "$DB_URL"
postgres://db.example.local:5432/mydb
saira@ubuntu:~$ echo "$DB_USER"
labuser
saira@ubuntu:~$ echo "$DB_PASSWORD"
labpass123
saira@ubuntu:~$ printenv | grep '^DB_'
DB_PASSWORD=labpass123
DB_USER=labuser
DB_URL=postgres://db.example.local:5432/mydb
saira@ubuntu:~$ _

```

3. Close and reopen terminal. Verify persistence by running one echo and the grep together — capture both in one screenshot:

echo "\$DB_URL"

printenv | grep '^DB_'

```
saira@ubuntu:~$ exit
```

```
Ubuntu 24.04.3 LTS ubuntu tty1
Hint: Num Lock on

ubuntu login: saira
Password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-87-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu  6 Nov 12:36:54 UTC 2025

System load:  0.0               Processes:            229
Usage of /:   81.4% of 9.75GB   Users logged in:     0
Memory usage: 9%               IPv4 address for ens33: 192.168.154.128
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

12 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

saira@ubuntu:~$ echo "$DB_URL"
postgres://db.example.local:5432/mydb
saira@ubuntu:~$ printenv | grep '^DB_'
DB_PASSWORD=labpass123
DB_USER=labuser
DB_URL=postgres://db.example.local:5432/mydb
saira@ubuntu:~$ _
```

Task 4 — System-wide environment variable, welcome script, and PATH

Goal: Add Class variable to `/etc/environment`, view `PATH`, create welcome script on Desktop, make executable, and add `PATH` entry in `~/.bashrc`. Capture grouped screenshots as applicable.

Steps:

1. View `/etc/environment`:

sudo cat /etc/environment

```
saira@ubuntu:~$ sudo cat /etc/environment
[sudo] password for saira:
PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin"
saira@ubuntu:~$
```

2. Show current `PATH` (single print) — this is not a `grep`, but keep it as a dedicated screenshot:

echo "\$PATH"

```
saira@ubuntu:~$ echo "$PATH"
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
saira@ubuntu:~$ _
```

3. Edit /etc/environment and add Class:

sudo vim /etc/environment

add line: Class="CC-<your_class_name>"

```
PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin"
Class="CC-<BSE-5B>
```

```
~
~
~
~
~
~
~
```

```
~
"/etc/environment" 2L, 136B written
saira@ubuntu:~$ sudo cat /etc/environment
```

```
PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin"
Class="CC-<BSE-5B>
saira@ubuntu:~$ _
```

4. Re-login or open a new shell and show Class and PATH together (grouped prints):
run echo \$Class and echo \$PATH together and capture in a single screenshot:

echo \$Class

echo "\$PATH"

```
saira@ubuntu:~$ source /etc/environment
saira@ubuntu:~$
saira@ubuntu:~$ echo $Class
CC-<BSE-5B>
saira@ubuntu:~$ echo "$PATH"
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
saira@ubuntu:~$ _
```

5. Create welcome script on Desktop and make it executable (capture the heredoc creation and chmod together in one screenshot if possible):

cat > ~/welcome <<'EOF'

#!/bin/bash

echo "Welcome to Cloud Computing \$USER"

EOF

chmod +x ~/welcome

```
saira@ubuntu:~$ cat > ~/welcome <<'EOF'
>
> #!/bin/bash
>
> echo "Welcome to Cloud Computing $USER"
>
> EOF
saira@ubuntu:~$
saira@ubuntu:~$ chmod +x ~/welcome
saira@ubuntu:~$ cd ~
saira@ubuntu:~$ ./welcome
Welcome to Cloud Computing saira
saira@ubuntu:~$
```

6. Run the script from Desktop using ./welcome (dedicated screenshot):

cd ~

./welcome

```
saira@ubuntu:~$ cd ~
saira@ubuntu:~$ ./welcome
Welcome to Cloud Computing saira
saira@ubuntu:~$ _
```

7. Add Desktop to PATH in ~/.bashrc (show the editor containing the PATH addition in one screenshot) and then source; capture the source and a single invocation of welcome (from home) together:

vim ~/.bashrc

add at end:

PATH=\$PATH:~/Desktop

export PATH

```

# colored GCC warnings and errors
#export GCC_COLORS='error=01;31:warning=01;35:note=01;36:caret=01;32:locus=01:quote=01'

# some more ls aliases
alias ll='ls -alF'
alias la='ls -A'
alias l='ls -CF'

# Add an "alert" alias for long running commands. Use like so:
# sleep 10; alert
alias alert='notify-send --urgency=low -i "${?} = 0" && echo terminal || echo error)' "${history|tail -n1|sed -e '\s/^s*[0-9]\*\s*//;s/[:&]\s*alert$//'\s*}'

# Alias definitions.
# You may want to put all your additions into a separate file like
# ~/.bash_aliases, instead of adding them here directly.
# See /usr/share/doc/bash-doc/examples in the bash-doc package.

if [ -f ~/.bash_aliases ]; then
    . ~/.bash_aliases
fi

# enable programmable completion features (you don't need to enable
# this, if it's already enabled in /etc/bash.bashrc and /etc/profile
# sources /etc/bash.bashrc).
if ! shopt -o posix; then
    if [ -f /usr/share/bash-completion/bash_completion ]; then
        . /usr/share/bash-completion/bash_completion
    elif [ -f /etc/bash_completion ]; then
        . /etc/bash_completion
    fi
fi

# add at the end:
# Lab 7 persistent DB variables

export DB_URL="postgres://db.example.local:5432/mydb"

export DB_USER="labuser"

export DB_PASSWORD="labpass123"

export PATH
export STUDENT_NAME="Saira Ejaz"

export STUDENT_ROLL_NUMBER="2023-BSE-057"

export STUDENT_SEMESTER="5th"
PATH=$PATH:~

-- INSERT --

```

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source ~/.bashrc

cd ~

welcome

```

saira@ubuntu:~$ source ~/.bashrc
Command 'ii' not found, but can be installed with:
sudo apt install ii
saira@ubuntu:~$
saira@ubuntu:~$ cd ~
saira@ubuntu:~$ welcome
Welcome to Cloud Computing saira
saira@ubuntu:~$

```

Task 5 — Block and allow SSH using ufw

Goal: Use ufw to deny and allow SSH then verify SSH connectivity changes from host. Save screenshots after each logical command/step; group related print checks when appropriate.

Steps:

1. Enable ufw and show status (group both commands in one screenshot if you run them together):

sudo ufw enable

sudo ufw status verbose


```
saira@ubuntu:~$ sudo ufw enable
Firewall is active and enabled on system startup
saira@ubuntu:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip
saira@ubuntu:~$ _
```

2. Deny TCP port 22 and show status (run deny and status numbered together and capture in one screenshot):

sudo ufw deny 22/tcp

sudo ufw status numbered

```
saira@ubuntu:~$ sudo ufw deny 22/tcp
Rule updated
Rule updated (v6)
saira@ubuntu:~$ sudo ufw status numbered
Status: active

      To      Action      From
      --      -
[ 1] 22/tcp    DENY IN     Anywhere
[ 2] 22/tcp (v6) DENY IN     Anywhere (v6)

saira@ubuntu:~$ _
```

3. From Windows host attempt to SSH (expected to fail) — capture the host-side SSH attempt in one screenshot:

ssh username@<server_ip>

```
PS C:\Users\saira> ssh saira@192.168.154.128
ssh: connect to host 192.168.154.128 port 22: Connection timed out
PS C:\Users\saira> |
```

4. Allow SSH back and reload, then show status (group allow, reload, status in one screenshot if run together):

sudo ufw allow 22/tcp

sudo ufw reload

sudo ufw status

```
saira@ubuntu:~$ sudo ufw allow 22/tcp
[sudo] password for saira:
Rule updated
Rule updated (v6)
saira@ubuntu:~$ sudo ufw reload
Firewall reloaded
saira@ubuntu:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)

saira@ubuntu:~$
```

5. From Windows host attempt SSH again (should succeed) — capture successful login in one screenshot:

ssh username@<server_ip>

```
PS C:\Users\saira> ssh saira@192.168.154.128
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-87-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri 7 Nov 05:13:19 UTC 2025

System load:  0.05      Processes:            231
Usage of /:   81.5% of 9.75GB   Users logged in:     1
Memory usage: 9%      IPv4 address for ens33: 192.168.154.128
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

12 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Thu Nov 6 13:22:52 2025 from 192.168.154.128
Command 'ii' not found, but can be installed with:
sudo apt install ii
saira@ubuntu:~$ |
```

Task 6 — Configure SSH key-based login from Windows host

Goal: Copy your public key from the Windows host into the Ubuntu server's ~/.ssh/authorized_keys to allow passwordless SSH. Save grouped screenshots for the client-side actions and the server-side edits/checks.

A. On Windows host (client) — group related client actions:

1. Generate ed25519 key pair (if needed) and show the generated files in one screenshot (run ssh-keygen and then list ~/.ssh):

ssh-keygen -t ed25519 -f ~/.ssh/id_ed25519 -C "lab_key"

```

PS C:\Users\saira> ssh-keygen -t ed25519 -f $env:USERPROFILE\.ssh\id_lab7 -C "lab_key"
Generating public/private ed25519 key pair.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in C:\Users\saira\.ssh\id_lab7
Your public key has been saved in C:\Users\saira\.ssh\id_lab7.pub
The key fingerprint is:
SHA256:9ps3bygthgVoDYjytDCbyBKPCcLcyPvLYHBLX29jPco lab_key
The key's randomart image is:
+--[ED25519 256]--+
|+ o . . |
|B=o.+ . . |
|=&.+ . = . . |
|Oo= + o = o |
|o.. . S= o . |
|o . . .E |
|. o . . o.. . |
| o . . +o= . |
| .o+ +. |
+-----[SHA256]-----+
PS C:\Users\saira>

```

ls -la ~/.ssh

```

PS C:\Users\saira> ls $env:USERPROFILE\.ssh

Directory: C:\Users\saira\.ssh

Mode                LastWriteTime         Length Name
----                -
-a-----         11/6/2025   9:33 PM             399 id_ed25519
-a-----         11/6/2025   9:33 PM              94 id_ed25519.pub
-a-----         11/6/2025   9:36 PM             399 id_lab7
-a-----         11/6/2025   9:36 PM              90 id_lab7.pub
-a-----         11/6/2025   9:31 PM              0 known_hosts
-a-----         11/4/2025  12:49 AM              98 known_hosts.old

PS C:\Users\saira>

```

2. Show the public key content (single screenshot):

type \$env:USERPROFILE\.ssh\id_lab7.pub

or on Git Bash: cat ~/.ssh/id_lab7.pub

```

PS C:\Users\saira> type $env:USERPROFILE\.ssh\id_lab7.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIBGp8b0w9tE6iZSgnrh/giy4ktubr5gyUy3X1RGe3spx lab_key

```

3. (Optional) Show known_hosts after first connection (single screenshot):

type \$env:USERPROFILE\.ssh\known_hosts

```

PS C:\Users\saira> type $env:USERPROFILE\.ssh\known_hosts
PS C:\Users\saira>

```

B. On Ubuntu server — group related server-side commands:

1. Inspect .ssh and authorized_keys before adding key (capture both outputs in one screenshot):

ls -la ~/.ssh

```
saira@ubuntu:~$ ls -la ~/.ssh
total 24
drwx----- 2 saira saira 4096 Nov  6 13:04 .
drwxr-x--- 24 saira saira 4096 Nov  6 12:54 ..
-rw----- 1 saira saira   0 Nov  3 14:13 authorized_keys
-rw----- 1 saira saira  444 Nov  6 13:04 id_ed25519
-rw-r--r-- 1 saira saira   89 Nov  6 13:04 id_ed25519.pub
-rw----- 1 saira saira  978 Nov  4 08:51 known_hosts
-rw-r--r-- 1 saira saira  142 Nov  4 08:51 known_hosts.old
saira@ubuntu:~$ _
```

cat ~/.ssh/authorized_keys || true

```
saira@ubuntu:~$ cat ~/.ssh/authorized_keys || true
saira@ubuntu:~$
```

2. Create ~/.ssh, set permissions, append the public key (manual method), and show the resulting authorized_keys — capture the mkdir/chmod/echo and the final cat in one screenshot:

mkdir -p ~/.ssh

chmod 700 ~/.ssh

```
saira@ubuntu:~$ mkdir -p ~/.ssh
saira@ubuntu:~$ chmod 700 ~/.ssh
saira@ubuntu:~$ echo "ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIM6BgSU9Ypup1aWT+G6cQ0w19e2mRDtGoAwIUch1Nwxg lab_key" >> ~/.ssh/authorized_keys
saira@ubuntu:~$ _
```

paste public key line from Windows client into the echo below (or use ssh-copy-id)

echo "ssh-ed25519 AAAA... yourpublickey ... comment" >> ~/.ssh/authorized_keys

chmod 600 ~/.ssh/authorized_keys

cat ~/.ssh/authorized_keys

```
saira@ubuntu:~$ chmod 600 ~/.ssh/authorized_keys
saira@ubuntu:~$ cat ~/.ssh/authorized_keys
ssh-ed25519 AAAA... yourpublickey ... comment
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIM6BgSU9Ypup1aWT+G6cQ0w19e2mRDtGoAwIUch1Nwxg lab_key
saira@ubuntu:~$ _
```

3. From Windows host test passwordless login (capture successful login in one screenshot):

ssh username@<server_ip>

```

PS C:\Users\saira> ssh saira@192.168.154.128
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-87-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu  6 Nov 13:21:54 UTC 2025

System load:  0.08               Processes:           237
Usage of /:   81.4% of 9.75GB    Users logged in:    1
Memory usage: 9%                IPv4 address for ens33: 192.168.154.128
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

12 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Thu Nov  6 13:00:38 2025 from 192.168.154.128
saira@ubuntu:~$ |

```

4. Also demonstrate explicit identity usage (single screenshot):

ssh -i ~/.ssh/id_ed25519 username@<server_ip>

```

saira@ubuntu:~$ ssh -i $env:USERPROFILE\.ssh\id_ed25519 saira@192.168.154.128
Warning: Identity file :USERPROFILE.sshid_ed25519 not accessible: No such file or directory.
saira@192.168.154.128's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-87-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu  6 Nov 13:21:54 UTC 2025

System load:  0.08               Processes:           237
Usage of /:   81.4% of 9.75GB    Users logged in:    1
Memory usage: 9%                IPv4 address for ens33: 192.168.154.128
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

12 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Thu Nov  6 13:21:55 2025 from 192.168.154.1
saira@ubuntu:~$ |

```

Exam Evaluation Questions

Q1: Quick Environment Audit

- Objective: Demonstrate you can inspect the current environment and extract a few key variables.
- Actions & evidence:
 - i. Run a single command to display environment variables and capture its output.

```
saira@ubuntu:~$ printenv

SHELL=/bin/bash
DB_PASSWORD=laboass123
CREDENTIALS_DIRECTORY=/run/credentials/getty@tty1.service
MEMORY_PRESSURE_WRITE=c29t2SgYMDAuMDAgMJAuMDAuMAA=
XDG_SEAT=seat0
PWD=/home/saira
LOGNAME=saira
XDG_SESSION_TYPE=ttty
SYSTEMD_EXEC_PID=1476
DB_USER=labuser
HOME=/home/saira
LANG=en_GB.UTF-8
LS_COLORS=rs=0:di=01:34:ln=01:36:mh=00:pi=40:33:so=01:35:do=01:35:bd=40:33:01:cd=40:33:01:or=40:31:01:mi=00:su=37:41:sg=30:43:ca=00:tw=30:42:ow=34:42:st=37:44:
e=01:32:*.tar=01:31:*.taz=01:31:*.arc=01:31:*.arj=01:31:*.taz=01:31:*.lha=01:31:*.lzh=01:31:*.lzm=01:31:*.tlz=01:31:*.txz=01:31:*.tzo=01:31:*.t7z=0
1:31:*.zip=01:31:*.z=01:31:*.dz=01:31:*.gz=01:31:*.lrz=01:31:*.lz=01:31:*.lzo=01:31:*.xz=01:31:*.zst=01:31:*.tzt=01:31:*.bz2=01:31:*.bz=01:31:*.tbz=01:31:*.tbz
2=01:31:*.t2=01:31:*.deb=01:31:*.rpm=01:31:*.jar=01:31:*.war=01:31:*.ear=01:31:*.sar=01:31:*.rar=01:31:*.alz=01:31:*.ace=01:31:*.zoo=01:31:*.cpio=01:31:*.7z=01:
31:*.rz=01:31:*.cab=01:31:*.wim=01:31:*.sum=01:31:*.dmg=01:31:*.esd=01:31:*.avif=01:35:*.jpg=01:35:*.jpeg=01:35:*.mjpg=01:35:*.mjpeg=01:35:*.gif=01:35:*.bmp=01:
35:*.pbm=01:35:*.pgm=01:35:*.ppm=01:35:*.tga=01:35:*.xbm=01:35:*.xpm=01:35:*.tif=01:35:*.tiff=01:35:*.png=01:35:*.svg=01:35:*.svgz=01:35:*.mng=01:35:*.pcc=01:35
:*.mov=01:35:*.mpg=01:35:*.mpeg=01:35:*.m2v=01:35:*.mkv=01:35:*.webm=01:35:*.webp=01:35:*.ogm=01:35:*.mp4=01:35:*.m4v=01:35:*.mp4v=01:35:*.vob=01:35:*.qt=01:35
:*.nuv=01:35:*.wmv=01:35:*.asf=01:35:*.rm=01:35:*.rmvb=01:35:*.flc=01:35:*.avi=01:35:*.fli=01:35:*.flv=01:35:*.gl=01:35:*.dl=01:35:*.xcf=01:35:*.xwd=01:35:*.yuv=
01:35:*.cgm=01:35:*.emf=01:35:*.ogv=01:35:*.ogx=01:35:*.aac=00:36:*.au=00:36:*.flac=00:36:*.m4a=00:36:*.mid=00:36:*.midi=00:36:*.mka=00:36:*.mp3=00:36:*.mpc=00:
36:*.ogg=00:36:*.ra=00:36:*.uav=00:36:*.oga=00:36:*.opus=00:36:*.spx=00:36:*.xspf=00:36:*.w=00:90:*.w=00:90:*.bak=00:90:*.crdownload=00:90:*.dpg=00:90:*.dpg=00:90:*.dpg
e-new=00:90:*.dpg-old=00:90:*.dpg-tmp=00:90:*.old=00:90:*.orig=00:90:*.part=00:90:*.rej=00:90:*.rpmnew=00:90:*.rpmorig=00:90:*.rpmsave=00:90:*.src=00:90:*.tmp
=00:90:*.ucf-dist=00:90:*.ucf-new=00:90:*.ucf-old=00:90:
MEMORY_PRESSURE_PATH=/sys/fs/cgroup/system.slice/system-getty.slice/getty@tty1.service/memory.pressure
INVOCATION_ID=3d9ed0f6c71e4a379cde66952d21b621
LESSCLOSE=/usr/bin/lesspipe %s %s
XDG_SESSION_CLASS=user
TERM=linux
LESSOPEN=| /usr/bin/lesspipe %s
USER=saira
SHLVL=1
XDG_VTNR=1
DB_URL=postgres://db.example.local:5432/mydb
XDG_SESSION_ID=1
XDG_RUNTIME_DIR=/run/user/1000
XDG_DATA_DIRS=/usr/share/ gnome:/usr/local/share:/usr/share:/var/lib/ snapd/desktop
HUSHLOGIN=FALSE
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/home/saira/Desktop
OBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
MAIL=/var/mail/saira
class=CC-BS-SSB
_/usr/bin/printenv
saira@ubuntu:~$
```

- ii. In the same terminal session, run three filters (one per line) to show values for PATH, LANG, and PWD, then capture a single screenshot showing the three outputs together.

```
saira@ubuntu:~$ echo "$PATH"
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/home/saira/Desktop
saira@ubuntu:~$ echo "$LANG"
en_GB.UTF-8
saira@ubuntu:~$ echo "$PWD"
/home/saira
saira@ubuntu:~$
```

Q2: Short-lived Student Info

- Objective: Show how temporary environment variables behave (session-scoped).
- Actions & evidence:
 - i. In one terminal, set three variables (STUDENT_NAME, STUDENT_ROLL_NUMBER, STUDENT_SEMESTER) using export — execute all three consecutively and capture them in one screenshot (show the commands executed).

```
saira@ubuntu:~$ export STUDENT_NAME="Saira"
saira@ubuntu:~$ export STUDENT_ROLL_NUMBER="12345"
saira@ubuntu:~$ unset STUDENT_ROLL_NUMBER
saira@ubuntu:~$
saira@ubuntu:~$ export STUDENT_ROLL_NUMBER="2023-BSE-057"
saira@ubuntu:~$
saira@ubuntu:~$ export STUDENT_SEMESTER="5th"
saira@ubuntu:~$
```

- ii. Still in the same session, print the three values with echo (grouped) and capture the outputs in one screenshot.

```
saira@ubuntu:~$ echo "$STUDENT_NAME"
Saira
saira@ubuntu:~$ echo "$STUDENT_ROLL_NUMBER"
2023-BSE-057
saira@ubuntu:~$ echo "$STUDENT_SEMESTER"
5th
saira@ubuntu:~$ _
```

- iii. Use a single `printenv|grep` command to list any `STUDENT_` variables and capture the result.

```
saira@ubuntu:~$ printenv | grep '^STUDENT_'
STUDENT_NAME=Saira
STUDENT_SEMESTER=5th
STUDENT_ROLL_NUMBER=2023-BSE-057
saira@ubuntu:~$ _
```

- iv. Exit that shell, open a fresh terminal, and show that the `STUDENT_` variables are not set (use `echo` and `printenv|grep` together) — capture in one screenshot.

```
saira@ubuntu:~$ echo "$STUDENT_NAME"
Saira
saira@ubuntu:~$ printenv | grep '^STUDENT_'
STUDENT_NAME=Saira
STUDENT_SEMESTER=5th
STUDENT_ROLL_NUMBER=2023-BSE-057
saira@ubuntu:~$
saira@ubuntu:~$ _
```

Q3: Make It Sticky (Persistence Check for Student Info)

- Objective: Demonstrate persistence of environment variables across sessions via shell configuration.
- Actions & evidence:
 - i. Edit `~/.bashrc` and append the three `STUDENT_*` exports. Capture a screenshot of the editor showing the new lines.

```
GNU nano 7.2 /home/saira/.bashrc

# some more ls aliases
alias ll='ls -alF'
alias la='ls -A'
alias l='ls -CF'

# Add an "alert" alias for long running commands.  Use like so:
# sleep 10; alert
alias alert='notify-send --urgency=low -i "${[ $? = 0 ]} && echo terminal || echo error)" "${history|tail -n1|sed -e 's/\s*$/\s*alert$/'}"

# Alias definitions.
# You may want to put all your additions into a separate file like
# ~/.bash_aliases, instead of adding them here directly.
# See /usr/share/doc/bash-doc/examples in the bash-doc package.

if [ -f ~/.bash_aliases ]; then
    . ~/.bash_aliases
fi

# enable programmable completion features (you don't need to enable
# this; if it's already enabled in /etc/bash.bashrc and /etc/profile
# sources /etc/bash.bashrc).
if ! shopt -oq posix; then
    if [ -f /usr/share/bash-completion/bash_completion ]; then
        . /usr/share/bash-completion/bash_completion
    elif [ -f /etc/bash_completion ]; then
        . /etc/bash_completion
    fi
fi

# add at the end:
# Lab 7 persistent DB variables

export DB_URL="postgres://db.example.local:5432/mydb"

export DB_USER="labuser"

export DB_PASSWORD="labpass123"
PATH=$PATH:~/Desktop

export PATH
export STUDENT_NAME="Saira Ejaz"

export STUDENT_ROLL_NUMBER="2023-BSE-057"

export STUDENT_SEMESTER="5th"

-
[ Wrote 131 lines ]
⌘ Help ⌘ Write Out ⌘ Where Is ⌘ Cut ⌘ Execute ⌘ Location ⌘ Undo ⌘ Set Mark ⌘ To Bracket ⌘ Previous
⌘ Exit ⌘ Read File ⌘ Replace ⌘ Paste ⌘ Justify ⌘ Go To Line ⌘ Redo ⌘ Copy ⌘ Where Was ⌘ Next
```

- ii. Reload your shell config with a single command and then verify the three variables and show `printenv | grep '^STUDENT_'` — capture these verification outputs together in one screenshot.

```
saira@ubuntu:~$ source ~/.bashrc
Command 'ii' not found, but can be installed with:
sudo apt install ii
saira@ubuntu:~$ echo $STUDENT_NAME
Saira Ejaz
saira@ubuntu:~$ echo $STUDENT_ROLL_NUMBER
2023-BSE-057
saira@ubuntu:~$ echo $STUDENT_SEMESTER
5th
saira@ubuntu:~$ printenv | grep '^STUDENT_'
STUDENT_NAME=Saira Ejaz
STUDENT_SEMESTER=5th
STUDENT_ROLL_NUMBER=2023-BSE-057
saira@ubuntu:~$
```

- iii. Close and re-open a terminal and demonstrate the `STUDENT_NAME` variable is available (echo and `printenv grep` together) — capture in one screenshot.


```

Ubuntu 24.04.3 LTS ubuntu tty1

ubuntu login: saira
Password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-87-generic x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Fri  7 Nov 04:26:39 UTC 2025

System load:  0.0                Processes:            233
Usage of /:   81.5% of 9.75GB    Users logged in:     0
Memory usage: 9%                IPv4 address for ens3: 192.168.154.128
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

12 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Command 'ii' not found, but can be installed with:
sudo apt install ii
saira@ubuntu:~$ echo $STUDENT_NAME
Saira Ejaz
saira@ubuntu:~$ printenv | grep '^STUDENT_'
STUDENT_NAME=Saira Ejaz
STUDENT_SEMESTER=5th
STUDENT_ROLL_NUMBER=2023-BSE-057
saira@ubuntu:~$

```

Q4: Firewall Rules: Block and Restore Ping (ICMP)

- Objective: Demonstrate you can block ping (ICMP echo) traffic using ufw and then re-allow it; show effect from a client.
- Actions & evidence:
 - i. Enable ufw and capture the enable command and status together in one screenshot.

```

saira@ubuntu:~$ sudo ufw enable
[sudo] password for saira:
Firewall is active and enabled on system startup
saira@ubuntu:~$ sudo ufw status
Status: active

To      Action      From
--      -
22/tcp  ALLOW      Anywhere
22/tcp (v6)  ALLOW      Anywhere (v6)

saira@ubuntu:~$ _

```

- ii. Add a rule to block ping (ICMP echo) and show ufw status numbered in the same screenshot.
 - Suggested command example:
 - `sudo ufw deny proto icmp from any to any`

sudo ufw status numbered

```
saira@ubuntu:~$ sudo ufw deny proto icmp from any to any
ERROR: Unsupported protocol 'icmp'
saira@ubuntu:~$ sudo ufw status numbered
Status: active

      To      Action      From
      --      -
[ 1] 22/tcp    ALLOW IN    Anywhere
[ 2] 22/tcp (v6) ALLOW IN    Anywhere (v6)

saira@ubuntu:~$
```

- iii. From your Windows host (or another client), attempt to ping the server while the rule is active and capture the blocked/failing ping in one screenshot.

```
PS C:\Users\Lenovo> ping 192.168.154.128

Pinging 192.168.154.128 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
```

- i. Re-allow ping (ICMP) (or remove the deny rule) and capture the allow/reload/status sequence in one screenshot.

- Suggested command example:

sudo ufw allow proto icmp from any to any

sudo ufw reload

sudo ufw status

```
saira@ubuntu:~$ sudo ufw reload
[sudo] password for saira:
Firewall reloaded
saira@ubuntu:~$ sudo ufw status
Status: active

To      Action      From
--      -
22/tcp    ALLOW      Anywhere
22/tcp (v6) ALLOW      Anywhere (v6)

saira@ubuntu:~$
```

- ii. From the client, ping the server again and capture successful replies in one screenshot.

```
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),  
PS C:\Users\saira> ping 192.168.154.128
```

```
Pinging 192.168.154.128 with 32 bytes of data:
```

```
Reply from 192.168.154.128: bytes=32 time<1ms TTL=64
```

```
Reply from 192.168.154.128: bytes=32 time=1ms TTL=64
```

```
Reply from 192.168.154.128: bytes=32 time<1ms TTL=64
```

```
Reply from 192.168.154.128: bytes=32 time<1ms TTL=64
```

```
Ping statistics for 192.168.154.128:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
PS C:\Users\saira> |
```